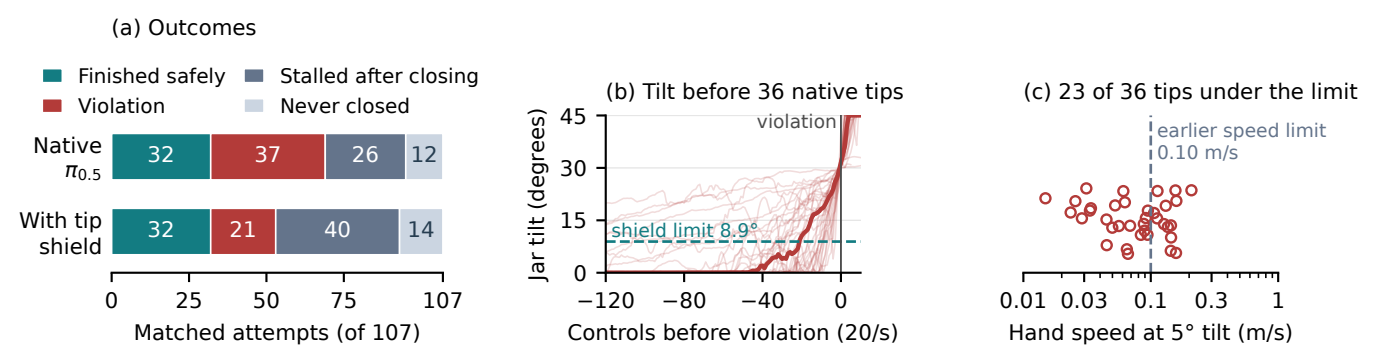
(a) Outcomes
Finished safely
Violation
Stalled after closing
Never closed
Native $\pi_{0.5}$
32
37
26
12
With tip shield
32
21
40
14
0
25
50
75
107
Matched attempts (of 107)
(b) Tilt before 36 native tips
violation
shield limit 8.9°
Jar tilt (degrees)
45
30
15
0
−120
−80
−40
0
Controls before violation (20/s)
(c) 23 of 36 tips under the limit
earlier speed limit
0.10 m/s
0.01
0.03
0.1
0.3
1
Hand speed at 5° tilt (m/s)